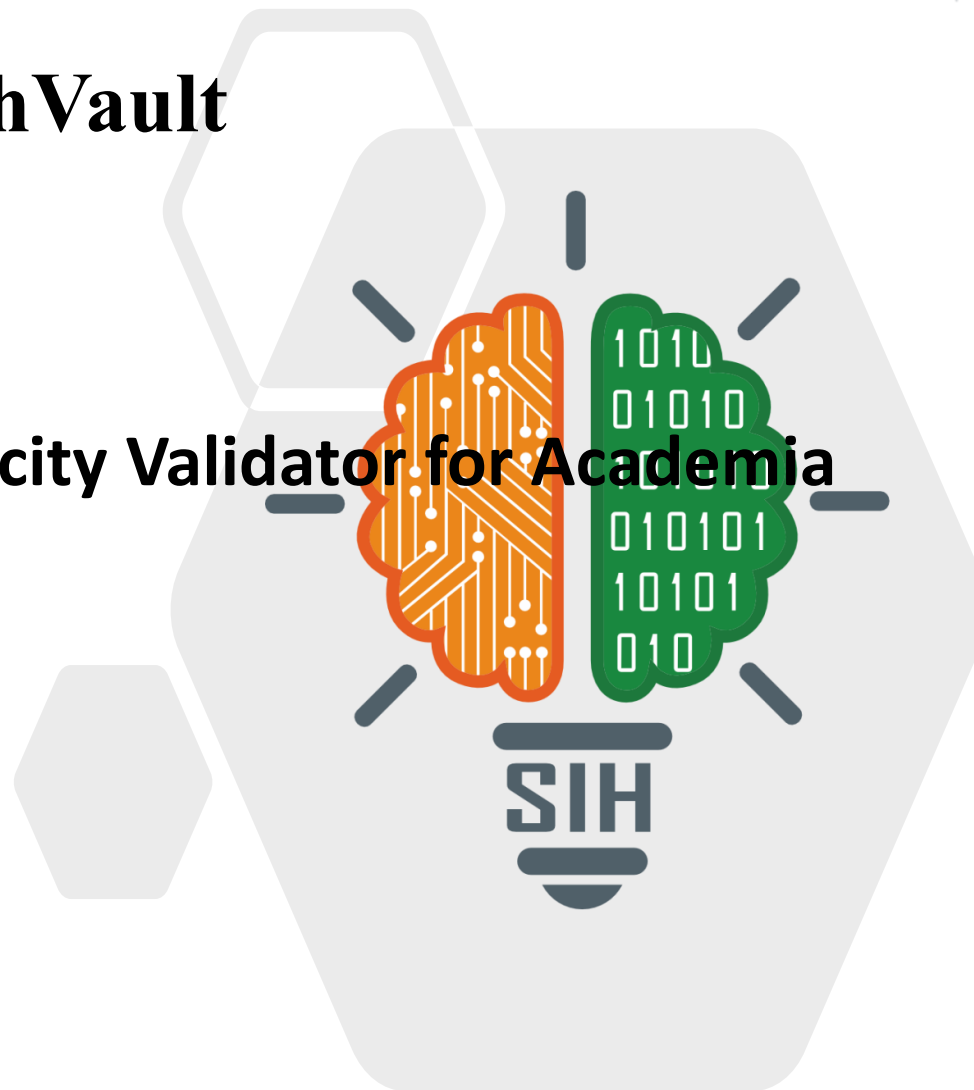


SMART INDIA HACKATHON 2026



HashVault

- Problem Statement ID –
- Problem Statement Title- Authenticity Validator for Academia
- Theme- Smart Education
- PS Category- Software
- Team ID-
- Team Name (Registered on portal)



Proposed Solution

Detailed Explanation

A unified **HashVault** platform for secure academic credential verification.

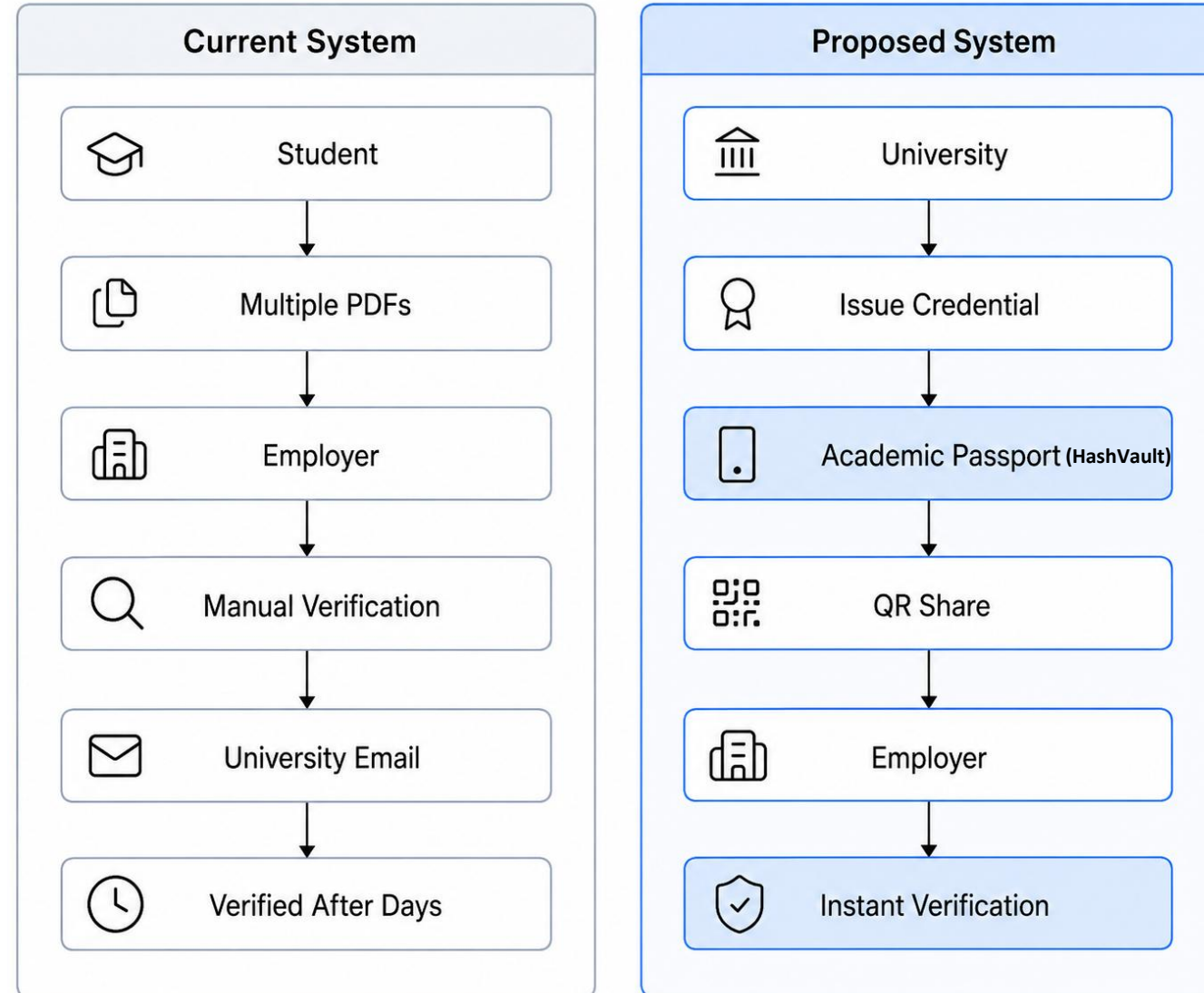
- **University** issues verified academic records.
- **SHA-256 Hash** generated for every document.
- **Academic Passport** created with **QR-based sharing**.
- **Employers** verify credentials instantly via **Verification Engine**.

How it Solves the Problem

- **Eliminates** manual document verification.
- **Instant QR-based** credential validation.
- **Tamper Detection** using Hash Verification.
- **One Verified Academic Identity** for every student.

Innovation & Uniqueness

- **Academic Passport** instead of multiple PDFs.
- **Cryptographic Hash + QR Verification**.
- **Future Integration:** DigiLocker, NAD & University APIs.



❖ Frontend

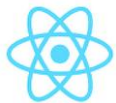
- React.js
- Responsive UI
- QR Scanner

❖ Backend

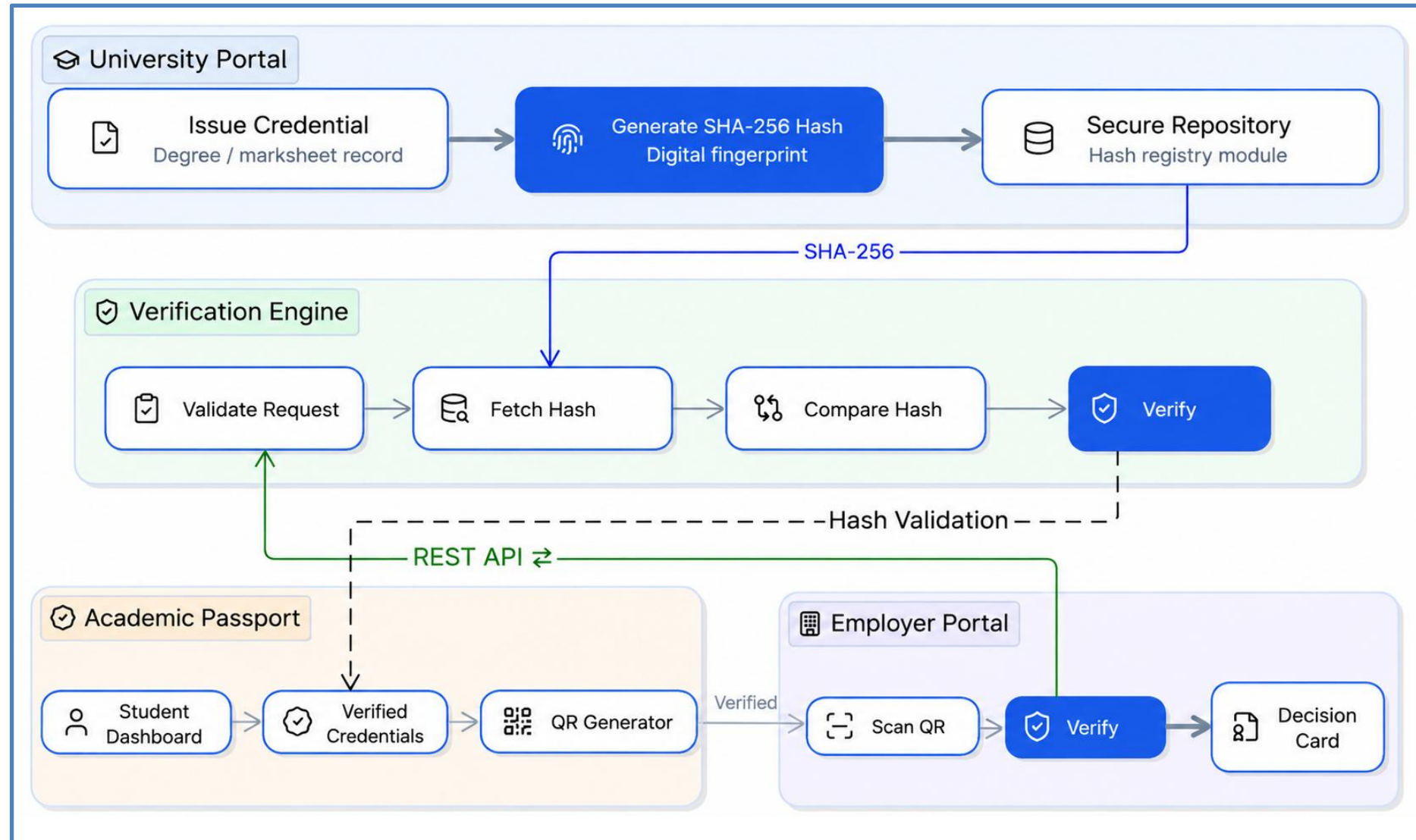
- Node.js + Express
- REST API
- MongoDB

❖ Security

- SHA-256 Hashing
- JWT Auth
- Tamper Detection



React





FEASIBILITY & VIABILITY :

- Uses Existing Infrastructure (Universities & DigiLocker Ready)
- Low-Cost Deployment (Cloud + REST APIs)
- Scalable Architecture (University → State → National)
- Fast Verification (<30 sec)



ECONOMIC FEASIBILITY

- Low Deployment Cost (Cloud-based solution)
- Minimal Maintenance (Automated verification workflow)
- Scalable Investment (University State National)
- Reduces Manual Verification Cost



OPERATIONAL FEASIBILITY

- Easy for Students, Universities & Employers
- 30-Second Verification Process
- No Training Required
- Paperless Academic Credential Management



CHALLENGES & SOLUTIONS

- University API Unavailable → Manual Upload + Hash Registry
- Legacy Records → Batch SHA-256 Hash Generation
- Fake Documents → Hash Comparison Engine
- Network Failure → Secure Cache + Retry Mechanism



KEY ADVANTAGES



Instant
QR Verification



Tamper-Proof
Credentials



Employer-Ready
Verification



Reduced
Background
Verification Time

❖ IMPACT & BENEFITS Of the Solution

Academic Impact

- **Lifetime Digital Academic Passport** for every student.
- **Instant & trusted credential verification** across institutions.

Employer Impact

- **Reduce hiring verification time** from days to seconds.
- **Eliminate manual certificate verification** and the risk of fraud.

University Impact

- **Secure digital credential issuance** with SHA-256 hashing.
- **Reduced administrative workload** through automated verification.

Economic & Future Impact

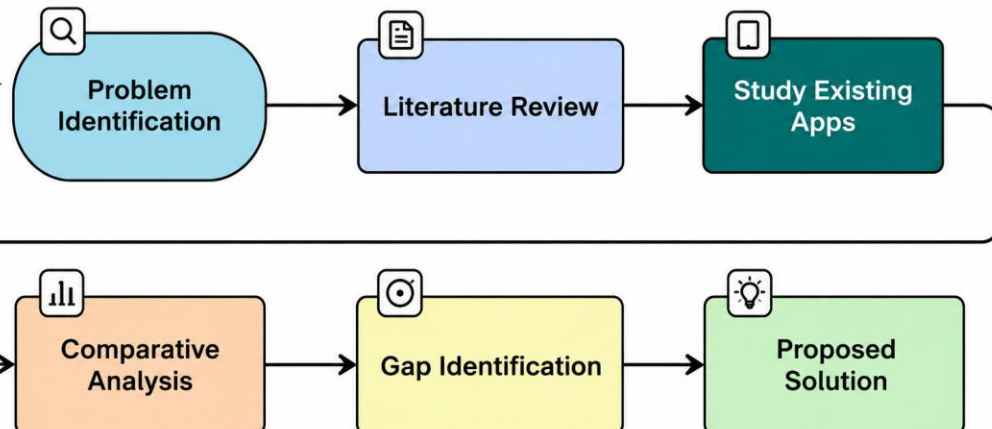
- **Low-cost, scalable solution** for universities nationwide.
- **DigiLocker-ready platform** with future AI & API integration.

❖ Comparison with Existing Systems

Feature	HashVault	DigiLocker	LinkedIn
Academic Passport	✓	×	×
QR Verification	✓	×	×
SHA-256 Hash Validation	✓	⚠	×
Employer Verification	✓ Instant	Limited	×
University API Integration	✓	✓	×
GitHub Integration	✓	×	⚠ Profile Only
Internship Verification	✓	×	×
Tamper Detection	✓	⚠	×
Verification Time	< 30 sec	Minutes	Manual
Fraud Detection	✓	×	×
REST API Ready	✓	✓	✓
Paperless Verification	✓	✓	×

Research & Best Practices (with Links)

- **Academic Bank of Credits (ABC)**
<https://www.abc.gov.in/>
- **National Academic Depository (NAD)**
<https://nad.digilocker.gov.in/>
- **NIST – SHA-256 Cryptographic Hash Standard**
<https://csrc.nist.gov/projects/hash-functions>
- **Sharma, S. (2026). DIGI-VAULT: A Centralised Digital Platform for Student Activity Records.**



Smart India Hackathon demo

The authenticity layer for academic credentials

Every degree, marksheet and internship certificate is hashed with SHA-256, digitally signed by the issuing university, and verifiable by any employer in under a second.

[Start issuing →](#) [Verify a passport](#)



University Issuer Portal

Create student records, upload degrees and certificates, auto-generate SHA-256 hashes and digital signatures.

[Open portal →](#)



Student Portal

A clean dashboard with verification status, certificates, GitHub identity, passport ID and a shareable QR code.

[Open portal →](#)



Employer Portal

Scan a QR or enter a passport ID, call the university verification API and read the decision card instantly.

[Open portal →](#)

Prototype Demo Link: [HashVault](#)